

Coupling the assessment of indoor environmental quality and cognitive performance in Building Information Modelling with integral indicators

Suzana Domjan^{a,*}, Rok Fink^b, Sašo Medved^a

^a Faculty of Mechanical Engineering, University of Ljubljana, Aškerčeva 6, 1000 Ljubljana, Slovenia

^b Faculty of Health Sciences, University of Ljubljana, Zdravstvena pot 5, 1000 Ljubljana, Slovenia

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ABSTRACT

Nowadays we spend most of time indoors and therefore building planners must provide best possible indoor environmental quality (IEQ) for living and working in the most energy efficient way. While living comfort can be predicted with the set of standardized physical variables, the evaluation of productivity is still not included in building design by default. The aim of the study is to present i) the experimental evaluation of performance of mental work of test subjects (age between 22 and 63 years) at controlled indoor environment in four IEQ domains – thermal, indoor air quality, lighting and noise exposure; ii) the statistical evaluation of three tests of mental work performed by test subjects (d2, arithmetical, Baddeley test), (iii) the method for ranking of IEQ based on physical variables and ranking the impact of physical variables on productivity of mental work as a basis for developing integral single score metric, (iv) a procedure to incorporate the integral single score metrics into dynamic modelling within the selected BIM tool. No statistically significant differences in scores of tests of mental work regarding to the gender and age of test subjects was found, while impact on productivity was found even if indoor environment conditions were maintained within the range of standardized quality classes (I–IV). Case study shown that indoor environmental quality metric significant increases by implementing energy efficiency measures, while smart operation of building technical system is needed to establish high productivity of mental work as well, those justifying the multi-approach of building design.

1. Introduction

Since we spend over 80 % of our lives indoors, it is understandable that research on indoor environmental quality (IEQ) is a common topic among researchers. While research on perceived IEQ prevails [1], an important part of the studies is dedicated to the evaluation of the performance of mental work. While IEQ can be measured with a broad range of indicators, including non-physical indicators such as cleaning and maintenance or the layout of furniture [2], it is common to both metrics that they are based on physical variables of the indoor environment conditions in four IEQ domains, i.e., thermal comfort, indoor air quality, lighting environment and acoustic and noise exposure [3]. While the perceived IEQ can be predicted with a set of standardized physical variables [4] that are ranked in quality classes (I for “top” to IV for “appropriate only for shorter occupant times”), the evaluation of IEQ impact on productivity, despite its importance, is still not introduced into building design by default. Nevertheless, building designers are not only responsible for the best possible living environment, but also for the

energy efficiency of the buildings. Since the building sector is responsible for 38 % of final energy use, it was agreed that buildings (new and renovated) should be transformed into zero-emission buildings in the near future [5]. With current building design practice, it is still not common to simultaneously address all those challenges. To approach such a multi-performance evaluation of the building-assessment process, a focus on the following research challenge must be made:

- Definition of the prediction models of a specific domain's indoor environmental quality and productivity. Most often, non-linear multi-parametric models are developed based on domain-specific physical variables [6]. While such models are statistically correct, they are not easily interpreted by building designers. Ranking the IEQ as well as productivity with discrete values will be a useful alternative, taking into account that such rankings are already introduced in a standardized evaluation of the indoor environmental quality [4]. As concluded by another researcher, the challenge is the lack of a quantitative model to assess the productivity of mental work [7,8].

* Corresponding author.

E-mail address: suzana.domjan@fs.uni-lj.si (S. Domjan).

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Nomenclature		θ_{mrt}	meat radiant temperature (°C)
Abbreviations		θ_{op}	operative temperature (°C)
BEM	Building Energy Modelling	R_{PL}	rank of productivity loss (–)
BIM	Building Information Modelling	s	standard deviation (unit)
DOE	Design of Experiment	T_1, T_2, T_3	test scores (%)
IEQ	Indoor Environmental Quality	UER	Euclidian distance (–)
T1, T2, T3	tests of productivity of mental work	w, u, z	weight factor of test scores (–)
Variables		Indexes	
CCT	correlated colour temperature of light (K)	1, 2, 3	test of mental work
CO ₂	concentration of CO ₂ (ppm)	avg	average
D	duration (of IEQ class) (h/period)	F	female
ΔPL	productivity loss (%)	IAQ	IEQ domain of indoor air quality
E_p	illumination of work plane (lx)	L	IEQ domain of light comfort
HI	humidity index (–)	M	male
IEQ _{INDEX}	single score metric of IEQ (–)	N	IEQ domain of noise exposure
φ_i	relative air humidity (%)	n	normalized (test scores)
L_p	sound pressure level (dB(A))	per	personal, of person
PL-IEQ _{INDEX}	single score metric of productivity of mental work (–)	T	IEQ domain of thermal comfort
PMV	Predicted Mean Vote (Fanger) (–)	w	weighted
θ_i	air temperature (°C)	x	IEQ domain (general)

- Definition of multi-domain IEQ and productivity metrics. The purpose of a multi-domain evaluation of IEQ and productivity is to i) combine the impact of several indoor environment domains. From 2 to 12 as follows from [1], ii) to evaluate the cross-dependency of the domains' impact (on IEQ and productivity) and iii) to define the weight factors of IEQ domains that indicate how the importance of the impact of each of the indoor environment domains on the integral metric of IEQ and productivity; taking into account that standardized evaluation of IEQ based on quality classes (I–IV) are accepted as normative, the evaluation should focused on the four above-mentioned IEQ domains. Research on a multi-domain evaluation of IEQ was presented by Schweiker et al. [9]. The authors reported that in approximately half of 219 articles it is reported that IEQ domains have no mutual influence on the perceived IEQ. In another review article about the combined effect of the IEQ domain [10] it is concluded that the cross-relationship between environmental variables still needs to be refined. A literature overview found that indicators/domain-weighting factors are commonly defined by a statistical correlation of the average vote of the perceived physical variables [11] or the productivity scores.
- Coupling the IEQ and productivity evaluation models into the dynamic modelling of building response. A determination of the IEQ, productivity and energy efficiency of buildings requires dynamic modelling, which can be a part of the building information modelling (BIM) process, since many BIM tools include a sub-level building energy modelling (BEM) module. Therefore, the IEQ and productivity metric can be simultaneously coupled with the dynamic modelling of buildings as part of the BIM process. This will increase the awareness and professionalism of building planners to a higher level, while involving all professionals will alleviate the problem that experts of different profiles emphasize different indoor environment domains [12]. As pointed out in [1], the presentation of results must be easy to understand by the building designer as well as the building users, preferably as integral, single score metric. The results of such an advanced building modelling could be to upgrade the buildings' Energy Performance Certificates with indoor environmental quality and a productivity-performance metric [13,14].

The aim and novelty of the presented study can be seen in: (i) the presentation of an experimental evaluation of the performance of mental

work by test subject at all stages of their professional career (aged 22–63 years) in a controlled indoor environment in four IEQ domains; ii) statistical evaluation of different tests of mental work that was the basis for developing the weighting factors of scores of tests, while the standard deviation of the results was included as a normalization factor in the test results to emphasise the personal productivity performance; (iii) presentation of the ranking of IEQ variables' impact on the productivity of mental work as a basis for developing an integral single-score productivity-loss metric, (iv) introduce a procedure to incorporate the integral indices of the IEQ and the productivity of mental work as an integral, single-score metric in dynamic modelling within the BIM tool by exchangeable macros, (v) the presentation of case studies to evaluate the IQE and productivity with an integral metric, with a focus on passive and free techniques to utilise on-site environmental energy sources, including smart controlling, the need to confirm/justify the need for a simultaneous evaluation of IEQ, productivity and the energy efficiency of buildings.

2. Experimental investigation of productivity of mental work in a controlled indoor environment

2.1. Test subjects and tests of cognitive abilities

The performance of the mental work was measured using cognitive tests undertaken by 46 subjects. Among them, 55 % were male and 45 % were female, aged between 22 and 65 years (representing the working population). For the statistical analyses, the test subjects, all actively working at the time, were divided into three age groups according to the assessed stage of their professional career, i.e., younger than 36 years, between 36 and 55 years, and older than 55 years. The test subjects were volunteers, selected from an online application, and were not involved in the education process or in a business relationship with the researchers. They did not receive any financial compensation for taking the tests. The choice of subjects by gender and age was tested with a Chi² test, which showed that there was no statistically significant correlation between these personal characteristics ($Chi^2 = 2.98, p = 0.255$). The tests were performed almost equally during periods of heating and cooling of the building.

The performance of mental work depends on visual observation and a personal capability to manage the cognitive process. The cognitive

process in the narrower sense is based on the cognitive functions [15,16], i.e., the perception and processing of information and the storage of information, attention, learning, which leads to the performance of executive functions (problem solving). In our study “paper-and-pencil” tests were used for cognitive ability and all the test subjects performed three tests (referred to as T1, T2, T3) that describe a person’s abilities in relation to the following aspects:

- T1 – visual perception and concentration were tested with the d2 test [17,18]; test consists of 14 lines with 47 characters each;
- T2 – numerical abilities were tested with arithmetic tests [3,19] designed for solving simple mathematical operation [20]; the tests required the subjects to mark pairs of two adjacent digits with a sum equal to 10; test consists of 25 lines with 32 numbers in each line;
- T3 – logical reasoning with the Baddeley test [19,21], a simple test of logical reasoning whose success has been shown to depend on environmental conditions; test consists of 83 logical questions.

From the range of indicators found in the literature [19,22], the following were used in our study: solving speed (Speed) and concentration (Conc).

2.2. Experimental IEQLab and indoor environment variables

The tests of mental work performance were carried out in an office-like room, which is shown in Fig. 2. The room consists of a working plane, artificial window, biophilia and acoustic elements, and was equipped with smart, controlled mechanical systems for cooling and heating the air, mechanical ventilation with heat recovery, lighting and the generation of noise. The acoustic elements were designed to provide a reverberation time of 1.2 s at low frequencies (<2000 Hz) to 0.9 s at high frequencies of sound (>6000 Hz). All the mechanical devices/systems can maintain the physical variables of the indoor environment in four domains (Fig. 1), i.e., thermal comfort, indoor air quality, lighting and exposure to noise, within the range of indoor environmental quality classes I to III, with occasional class IV for statistical reasons, as defined in the technical standard [4]. A 65" OLED display was installed as an artificial window, and each test subject could choose (during the initial adaptation phase) the window view that was most comfortable for him/her. A sound amplifier and two loudspeakers were used to simulate exposure to digital noise. The measuring devices were an Ahlborn IEQ for thermal comfort variables (Fig. 2, right), an Ahlborn FYAD 00 CO2

for IAQ, a UPRtek MK350N Premium spectrometer for illumination and correlated colour temperature (CCT) measurements, and an iMM-6 Dayton Audio for sound-pressure-level measurements. The sensors for the thermal comfort variables were placed perpendicular to the breathing zone of the test subjects and at a distance of 1.1 m. Any air pollutants were well mixed due the operation of the mechanical ventilation device. The data were recorded using an ALMEMO 710 data logger.

The indoor environment variables were controlled in two different ways:

- The quasi-controlled variables, which include thermal comfort and air quality, in our case the air temperature θ_a , mean radiative temperature θ_{mrt} , operative temperature θ_{op} , air humidity ϕ_i , air velocity v_i , and the CO₂ concentration, were random during the tests, but kept within the required IEQ classes [4]. Two combined thermal comfort indicators were determined on the basis of the measured data: the operative temperature $\theta_{op} = a_1 \times \theta_i + (1 - a_1) \times \theta_{mrt}$ with a_1 chosen to be equal to 0.5, and a humidity index $HI = (1.8 \times \theta_i + 32) - (0.55 - 0.0055 \times \phi_i) \times (1.8 \times \theta_i - 26)$ [23]. Fig. 3 shows the average values of these variables during each test of mental work performance within the experiment.
- The fully controlled variables are specified in pairs of two variables of the lighting and noise-exposure domain with discrete values: illuminance E_p (300 lx, 550 lx, 1500 lx; $\Delta E_p \pm 50$ lx) and correlated colour temperature CCT (2400 K, 4700 K, 6500 K; $\Delta CCT \pm 100$ K), sound-pressure level L_p (45 dB(A), 55 dB(A), 65 dB(A); $\Delta L_p \pm 2$ dB (A) and spectrum of noise (white noise, traffic noise, office noise). The variables were changed for each of the tests and kept constant during the adaptation and test phases (Fig. 4). To increase the statistical reliability, 9 combinations of discrete values of the IEQ controlled variables were set by applying the design of experiments (DOE) method and an orthogonal Taguchi matrix L9 (4 parameters, 3 levels). At least 5 subjects performed the mental work test for each setup.

2.3. Design of mental work performance experiment

Test subjects performed the experiment in the same order of tests as shown in Fig. 4. The total duration of the experiment was approximately 52 min, while the time for solving the test was 32:40 min and consisted of a period of initial adaptation during which the test subjects were

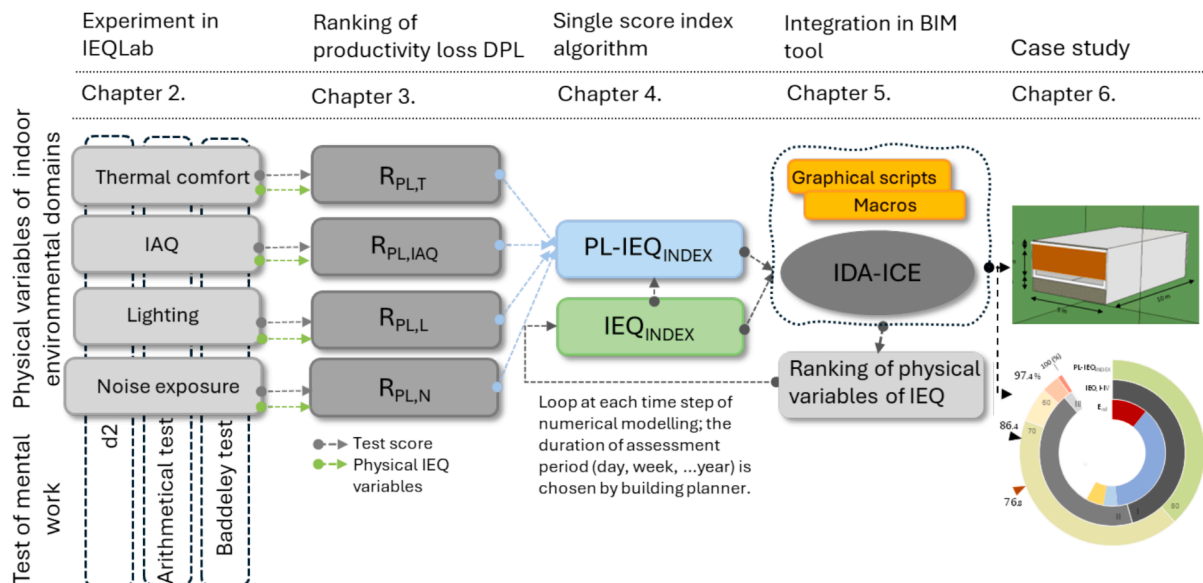


Fig. 1. Levels and tasks of the presented research.

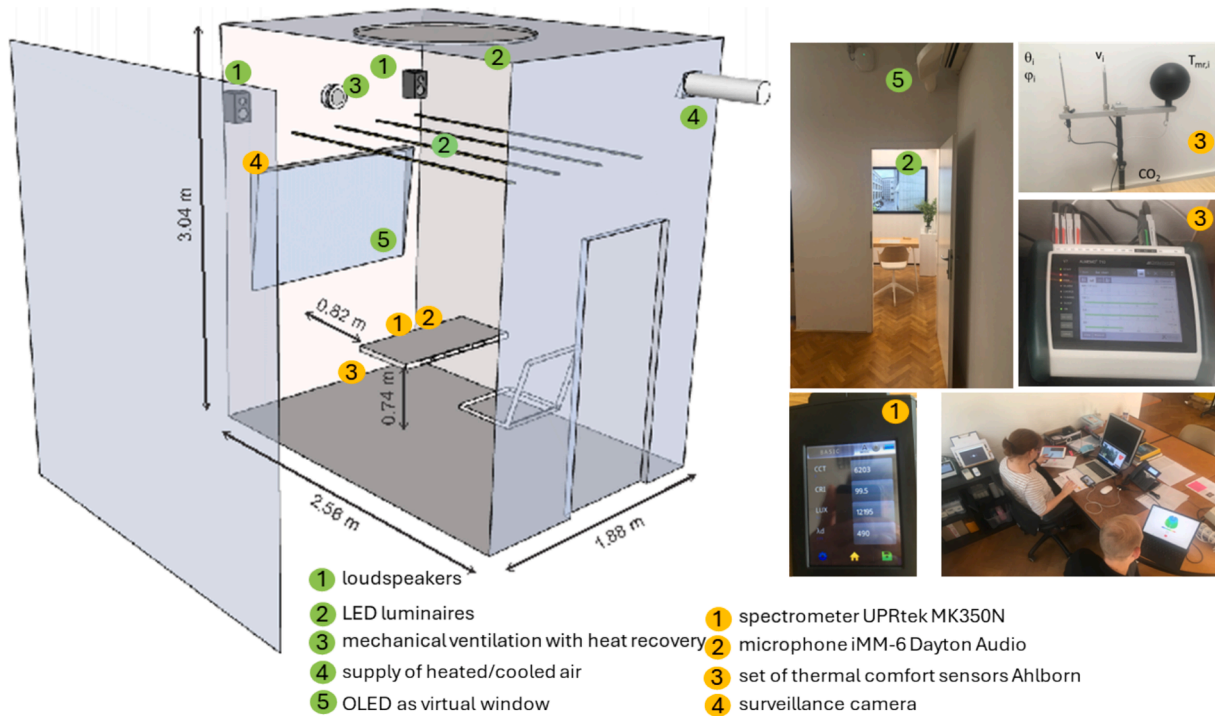


Fig. 2. IEQLab simulates office space and provides a controlled indoor environment in four domains (thermal comfort, IAQ, lighting and noise exposure).

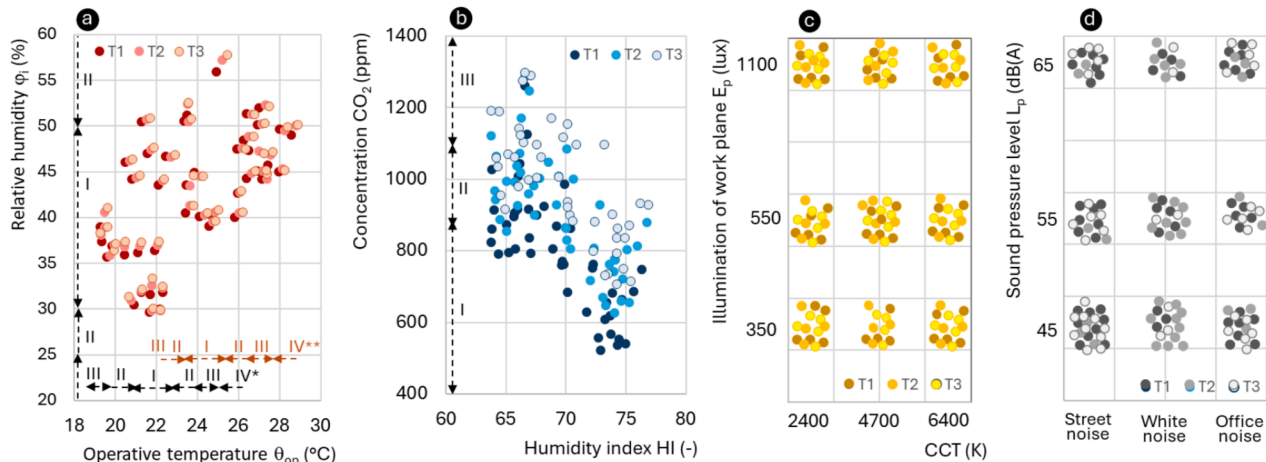


Fig. 3. Average values of the indoor environment variables of thermal comfort (a), indoor air quality (b) and controlled variables of light comfort domain (c) and noise exposure (d) during tests T1, T2, T3 (total 135 tests); IEQ quality classes (I–IV) (a, b) are defined according to [4]. Notes: * heating period; ** cooling period.

familiarized with the content of the tests and received basic general guidelines. During this period, the test subjects also selected the most pleasant scene through the window. This was shown on an artificial window for the entire duration of the experiment. Each test began with a relaxation phase using the same indoor environment for all the tests. The relaxation phase was followed by an adaptation phase, during which the indoor environment variables were set. After performing the test, the subjects answered a short questionnaire to express the perceived indoor environmental quality (phase Q). This procedure was repeated for each test, as shown in Fig. 4.

3. Experimental data on the productivity of mental work

The productivity of the mental work was measured with the concentration index (Conc). The number of correct answers was on a scale of 0 %–100 %. We skipped the other chosen index – the solving speed

(Speed), because we found that only a few test subjects did not manage to complete the test in the available time. The first result of the experiment that we want to highlight is the difference in the scores according to the type of test. The average scores were $T_1 = 57.6$ % (d2 test); $T_2 = 63.7$ % (arithmetic test) and $T_3 = 74.6$ % (Baddeley test). To develop the single-score productivity metric using all the available experimental results gathered under different indoor environment conditions, the weighting factors of the test scores were determined as the ratio between the average scores of each type of test and the average scores of all the tests: $w_{Ti} = T_{i,avg}/[avg(T_1, T_2, T_3)]$, with $i = 1–3$. The derived weighting factors $w_{T1} = 1.103$, $w_{T2} = 1.053$ and $w_{T3} = 0.874$ also showed that the developed single-score productivity metric will be suitable for the assessment of a wide range of mental work activities.

The second general conclusion comes from the score of the tests achieved by the different genders of the test subjects. The tests were performed by 78 male (n_M) and 57 female (n_F), and the average

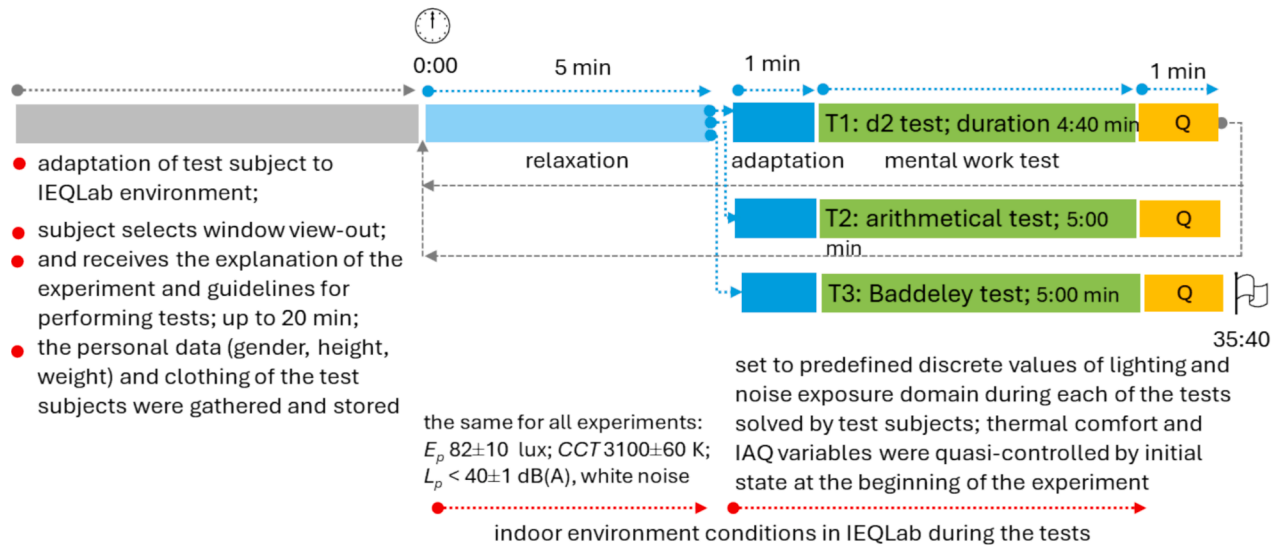


Fig. 4. Structure, tasks and duration of experiment. While the order of the tests was the same for all the test subjects, the indoor environment variables were different in each of the tests.

weighted scores for the test were very close: $T_{w,1,2,3,M} = 64.8\%$ and $T_{w,1,2,3,F} = 65.6\%$ (Fig. 5, left). Similarly, no statistically significant difference was found between the scores for the individual type of test: ($T_{1,M} = 56.6\%$, $T_{1,F} = 58.7\%$), ($T_{2,M} = 64.3\%$, $T_{2,F} = 63.0\%$), ($T_{3,M} = 73.4\%$, $T_{3,F} = 75.9\%$). The evaluation of the scores for the tests according to the age of test subjects was also evaluated. As shown in Fig. 5 (right) the mid-age group performed the test statistically significantly worse than the other two groups, but only because they performed worse on test T1 ($p = 0.02$), while no significant differences were found between the age groups in the scores for the T2 test ($p = 0.213$) and T3 (0.589).

Accordingly, we will not take gender and age as impacting personal characteristics when developing the prediction model for the single score of productivity.

3.1. Productivity of mental work under different indoor environment conditions

In this section an evaluation of the productivity of the mental work based on the experimental results of the test subjects according to the indoor environment during tests is presented. The ranking method was developed and ranks of productivity for the mental work were developed for each of the standardized assessment indoor environment domains, i.e., thermal comfort, indoor air quality, lighting environment

and exposure to noise, as presented in Fig. 1.

3.1.1. Impact of the thermal indoor environment on the productivity of mental work

Based on the measured variables of the indoor thermal environment, two combined indicators were determined in our study, i.e., the humidity index HI [23] and the predicted mean vote PMV (known as the Fanger's vote of indoor thermal environment) [24]. The first one combines the impact of the air temperature θ_i and the relative humidity ϕ_i ; the second one combines four physical variables (θ_i , θ_{mrt} , v_i , ϕ_i) and two personal characteristics, i.e., the activity/metabolism (met) and the thermal resistance of clothing (clo). Fig. 6 presents the scores achieved by the test subjects for different average HI index during the tests, as well as the trendline curves (a, b, c), while graph (d) shows the weighted scores of the tests. It can be concluded that the weighting factors $w_{T2,T2}$, T_3 can evaluate the different forms of mental work. From Fig. 6 (d) the highest productivity can be achieved when the humidity index HI is kept in the range 68–72. These correspond to the values of θ_i and ϕ_i as presented in Fig. 6 (d) for the heating and cooling periods.

The Perceived Mean Vote (PMV) predicts how a “standard” person perceives the indoor thermal environment. Such a person has standardized personal characteristics, such as height, weight and skin surface [25]. The clo adjusted to the yearly season and when performing sedentary work in an office building, has metabolic rate of 1.2 met.

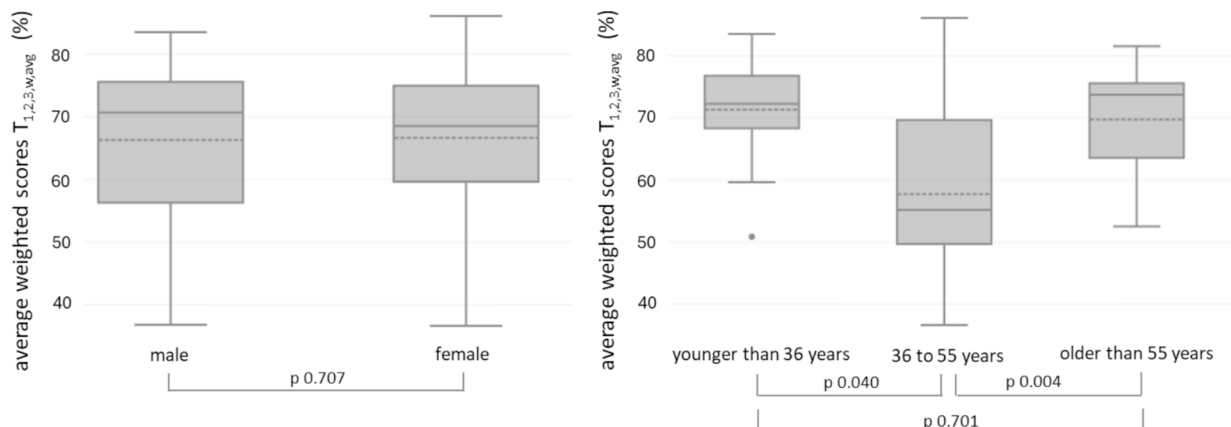


Fig. 5. Statistical evaluation of the weighted scores for the tests according to gender (left) and age group (right).

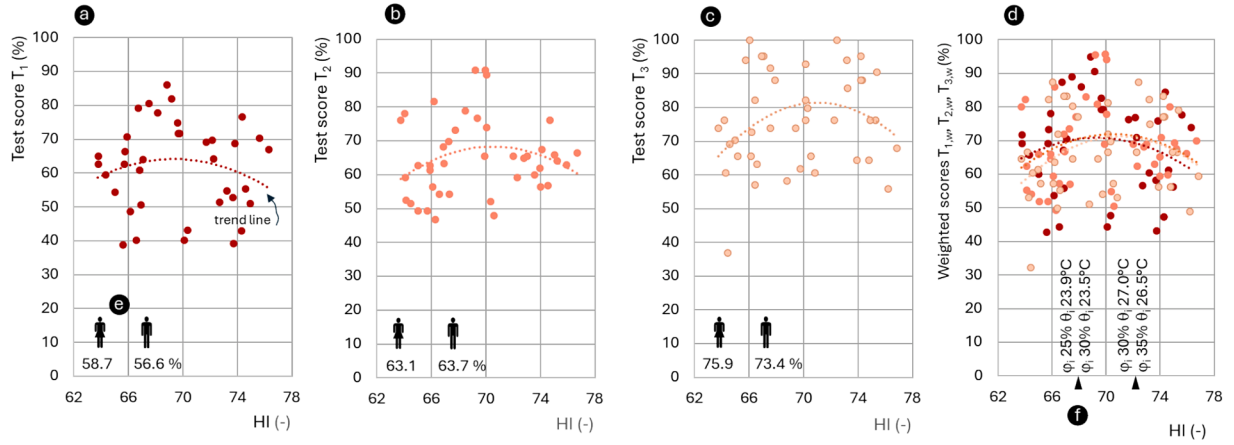


Fig. 6. Raw scores (unweighted) of mental work performance tests – T_1 (d2 test) (a), T_2 (arithmetic test) (b), T_3 (Baddeley test) (c) and weighted scores of tests $T_{1,w}$, $T_{2,w}$ and $T_{3,w}$ (d), average test scores according to the gender (e) and the Pareto values of indoor air conditions (heating/cooling period) at which productivity will be highest (f).

Later, we must define the amount of generated heat flux M (W) that the person's body must transfer to the surrounding environment. While we found that the clo of the test persons' clothing corresponded very well with the standardized seasonal values [26], i.e., during the heating period the average clo_M was 1.01 (standard deviation $s = 0.05$) and $clo_F = 0.96$ ($s = 0.14$), and during the cooling period the average clo_M was 0.6 ($s = 0.12$) and clo_F equals to 0.67 ($s = 0.23$), the personal characteristics of test subject differed a lot from the "standard" person. The interactive procedure was used to account for the actual personal characteristics when determining the metabolic rate [27]. For this reason, in the initial adaptation phase of the experiment (Fig. 4), the personal data of the test subjects, including gender, age, height, mass and empirical skin surface area, were collected. Using this data, the person-adjusted Predicted Mean Vote (PMV_{per}) was determined for each test subject and considered in the subsequent statistical analysis. Fig. 7 (1) shows the scores of the tests versus the test subjects' adjusted PMV_{per} . The trend curves indicate that the subjects perform the tests with the highest score at a PMV_{per} between -0.6 and -0.7 , which is a significant difference compared to the Fanger scale developed for a standardized person (where a PMV equal to 0 represents the highest value of thermal comfort). Our results can be compared with the studies of Roelofs [28] and Kosen & Tan [29], which all report the lowest productivity loss at $PMV = -0.5$.

The correlation between PMV and PMV_{per} is shown in Fig. 7 (2). Because the standardized thermal comfort quality classes are defined

according to PMV [4], i.e., class I of thermal comfort corresponds to $-0.2 < PMV < 0.2$, class II $-0.5 < PMV < 0.5$, class III $-0.7 < PMV < 0.7$ and class IV $-0.7 > PMV > 0.7$, as shown in Fig. 7(b), the correlation was used to determine the person-adjusted thermal comfort quality classes according to PMV_{per} as shown in Fig. 7(c). In the dynamic modelling of a building's response, the proposed ranges of classes of PMV_{per} should be adapted according to the season (Fig. 7(d)) for the heating and (d2) for the cooling season, following standardized PMV thermal comfort quality classes.

To determine the impact of the thermal indoor environment on the performance of mental work, the following procedure was developed:

- Tests are in four groups q ($q = I, II, III, IV$) according to the indoor environment thermal comfort quality class based on the PMV_{per} (Fig. 7, (2)) determined with indoor environment variables established during the test.
- The average values of the weighted scores of tests $T_{1;2;3,w}$ are determined for each group q of tests as $T_{w,avg,T,q}$, where the subscript T indicates the thermal comfort domain. Standard deviations of the scores of each group of tests $s_{T,q}$ are determined as well.
- Because from Fig. 6 and Fig. 7 it can be seen that the differences among the test scores for the same indoor environment conditions are evident regardless of the type of test, we concluded that the standard deviation of the scores for the tests should be included in

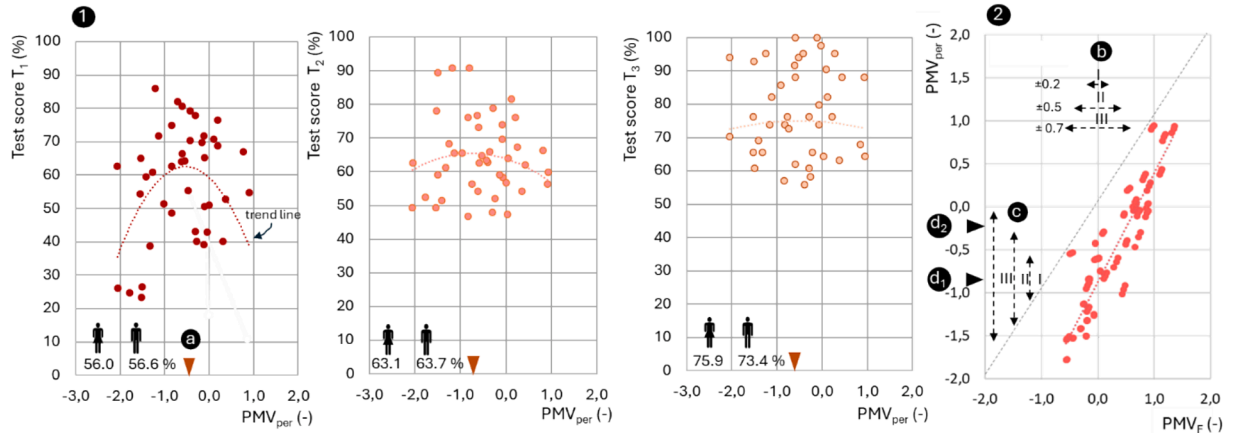


Fig. 7. Raw scores of tests of mental work performance (1); regardless of the type of the test, the higher scores were achieved at a person-adjusted Predicted Mean Vote (PMV_{per}) between -0.6 and -0.7 (a); correlation between PMV determined for test subjects as "standard" person and person-adjusted PMV_{per} calculated with average values of physical variables during each of the test (2).

the evaluation of the results as a normalization factor with the following relation (Eq. (1)):

$$T_{n,T,q} = \frac{1}{1 + \frac{s_{T,q}}{T_{w,avg,T,q}}} \cdot T_{w,avg,T,q} \quad (-), \quad (1)$$

where $T_{n,T,q}$ is the normalized average weighted scores of the tests performed at the q -th thermal comfort quality class of the indoor environment.

The normalized scores of tests $T_{n,T,q}$ are shown in Fig. 8 for the indoor environment thermal quality classes ($IEQ_{T,q}$) defined by PMV_{per} as shown in Fig. 7 (2). Although a statistically significant difference in the normalized score of the tests is only found between IEQ_T class I and class IV, the difference in the normalized scores can be found between other classes too and should be considered in the prediction model for the single-score productivity metric.

3.1.2. Impact of indoor air quality on the productivity of mental work

The impact of indoor air quality (IAQ) on the productivity of mental work was analysed according to the concentration of CO_2 above the outdoor environment, which was assumed to be a constant value of 320 ppm. The weighted scores of all the tests achieved at different average CO_2 concentrations during the test are shown in Fig. 9. In Fig. 9(a) the IAQ quality classes as defined in [4,30] are also shown and were considered by grouping the test results into q (I to IV) quality classes $IEQ_{IAQ,q}$. The average weighted test results $T_{w,avg,IAQ,q}$ and the standard deviation of the scores $s_{IAQ,q}$ are presented. Using the method described in Section 3.1.1, the test scores were normalized, and the results are shown in Fig. 9(b). The results indicate that statistically significant differences can only be found between IEQ_{IAQ} class I and class IV, although a statistically weak impact can be noticed between IEQ_{IAQ} classes II and III and class IV. This indicates that CO_2 concentrations up to 1000 ppm (IEQ_{IAQ} I–III) above the outdoor value have a low impact on mental work, yet concentrations above these values should be prevented as it obviously not only affects the perceived quality of the indoor environment, but also has an impact on mental work. These results are supported by the studies presented in [31–34].

3.1.3. Impact of the quality of lighting on the productivity of mental work

The test subjects performed mental work under artificial light. The productivity of mental work in the lighting domain was assessed by the weighted scores of the tests performed in different illuminations E_p of the work plane and the correlated colour temperature of the light CCT

emitted by the smart, controlled LEDs. Both variables were pre-selected according to the range of required values (E_p , office work) and the availability of the technology (CCT) in nine combinations g (Fig. 3(c)) selected by an orthogonal Taguchi matrix. The values that characterized the daylight conditions are set as: $E_p > 500$ lx, $CCT > 5500$ K. The weighted scores of the tests $T_{1,2,3,w}$ as well as the average values $T_{w,avg,L,g}$ and the standard deviations $s_{L,g}$ of the scores of the tests for each group g of test subjects achieved for the same controlled variables are shown in Fig. 10(a), while the normalized scores of tests $T_{n,IAQ,g}$ are shown in Fig. 10(b). The normalization method is described in Section 3.1.1.

We could confirm that a high productivity of mental work can be maintained by adjusting the CCT value of the emitted light to the required illuminance E_p . At lower illuminance levels, higher productivity will be achieved with warm or warm white light (CCT from 2400 K to 4700 K), while at higher required illuminance levels ($E_p > 550$ lx), cold white light (CCT 6400 K) is preferred. Under those conditions (E_p 550 lx, CCT 4700 K and E_p 1100 lx, CCT 6400 K), the standard deviations $s_{L,g}$ of the test results are the smallest. This confirms the results published in [35] and [36]. The findings in our study also confirm that the Kruithof relation [37] is an appropriate guideline for the design of lighting in buildings where mental work is performed, because for the best perceived indoor lighting quality, the highest productivity in mental work will be achieved (Fig. 10(c)). Although natural light in general is perceived as more pleasant than the artificial light used in our experiment [38], we follow the conclusion in [39], where the authors state that the performance of mental work did not decrease under the same lighting conditions when the work plane was illuminated with daylight or with artificial light.

3.1.4. Impact of exposure to noise on the productivity of mental work

As in the light domain, the productivity of mental work in the noise-exposure domain of IEQ was assessed by weighted and normalized scores of tests performed by test subjects exposed to predefined controlled noise variables, as shown in Fig. 3(d). The range and spectrum of the controlled variables were defined according to data from in-situ experiments in office buildings [40] and published experimental laboratory research [41] in which authors investigated the annoyance of different sound spectra in buildings. However, it should be noted that most noise-exposure studies were conducted with the, according to our findings, less annoying white noise [42].

The scores achieved by the subjects in our experiment are shown in Fig. 11. The results are presented as weighted scores $T_{1,2,3,w}$ for all the individual tests together with weighted average test scores $T_{w,avg,N,g}$ for groups performing the tests with the same noise-exposure variable, as well as with standard deviations $s_{N,g}$ of the test scores for each of the test groups (Fig. 11(a)). The normalized average test scores $T_{n,N,g}$ for each test group are shown in Fig. 11(b).

The results of the experiment confirm that the acoustic environment in which building users are exposed to constant background noise (white noise) increases their concentration for performing mental work when the sound-pressure level is kept in the range up to 55 dB(A). At higher levels of white noise exposure, productivity decreased and noise annoyance increased, as the standard deviation of the test scores increased with the intensity of the noise $L_{eq,A}$. The positive effect of background noise found in our study is also reported by Lu et al. [43], according to which the performance of work increased by 4.5 % when the subjects were exposed to 50 dB of white noise, compared to a quiet environment. Exposure to traffic noise and especially office noise has shown that these noise spectra affect the productivity much more than exposure to white noise. The unexpected increase in the performance of mental work observed at the highest sound-pressure level (65 dB(A)) can be explained by a stress condition. In this respect, our results are consistent with the research of Mehta et al. [44], in which the authors report that an acoustic environment with a constant background noise of 70 dB(A) (white noise) reduces feelings of anxiety and alienation more than noise exposure with an intensity of 50 dB(A). Our results are useful

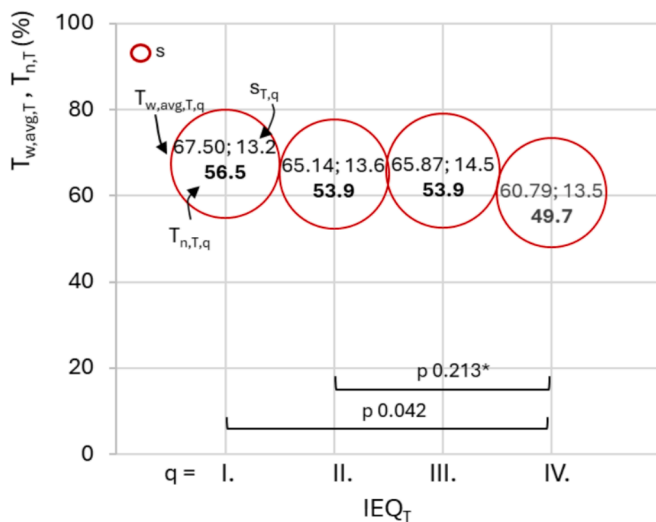


Fig. 8. Average weighted scores of tests $T_{w,avg,T,q}$, normalized scores $T_{n,T,q}$ and standard deviation of scores $s_{T,q}$ regarding the indoor environment thermal comfort class IEQ_T .

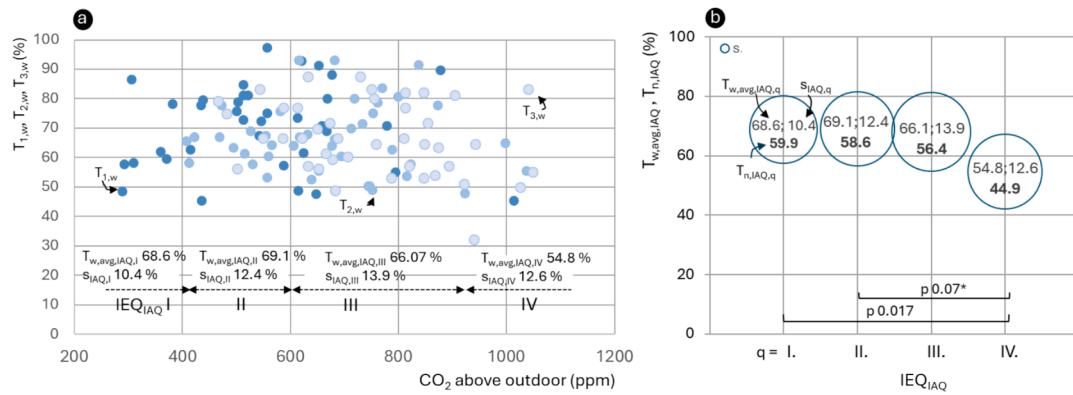


Fig. 9. Weighted scores of all tests of the performance of mental work $T_{1,2,3,w}$; average weighted scores of tests $T_{w,avg,IAQ,q}$ and standard deviation of scores $s_{IAQ,q}$ (left); normalized test scores $T_{n,IAQ,q}$ achieved indifferent indoor air quality classes $IEQ_{IAQ,q}$ (right).

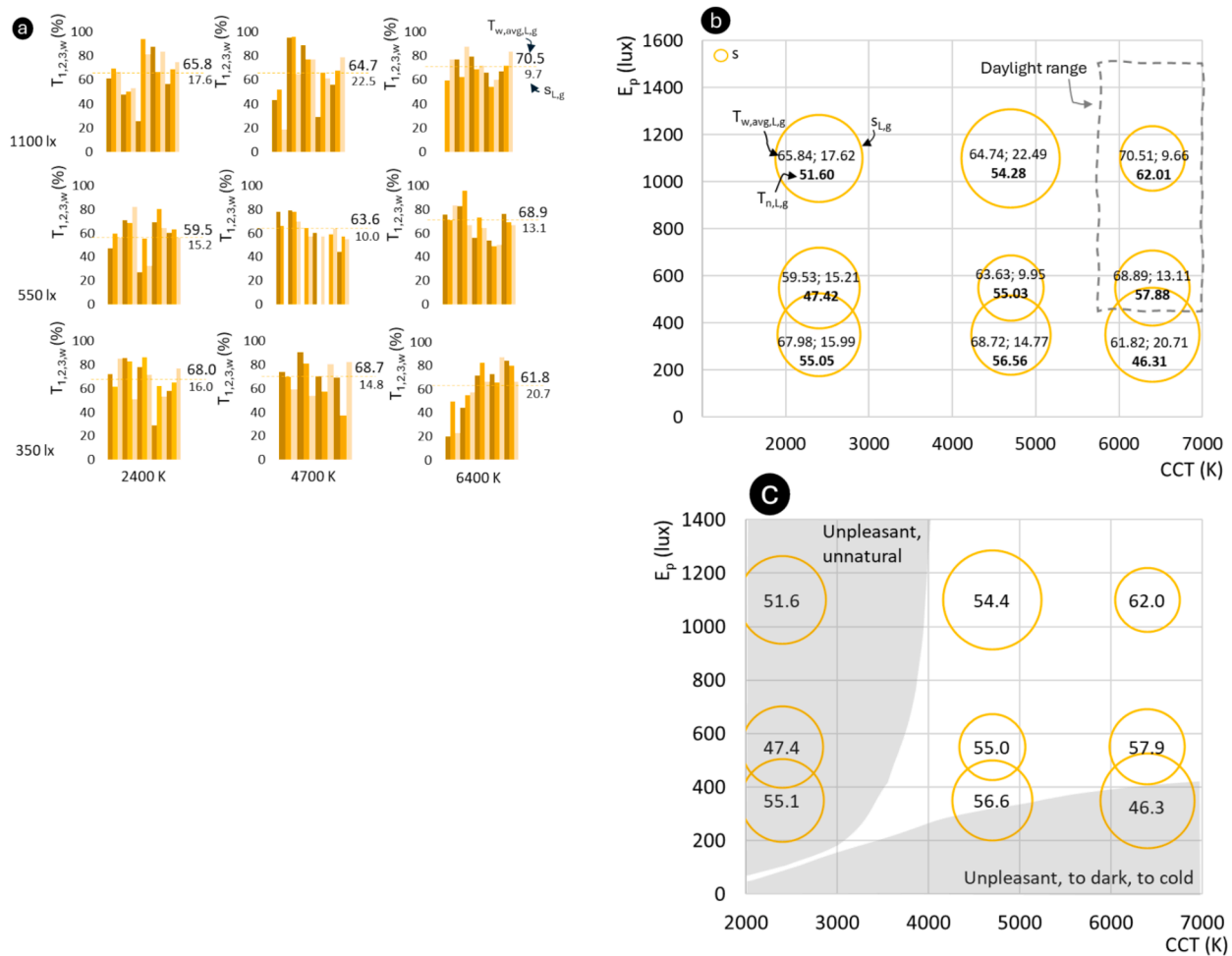


Fig. 10. Weighted test scores $T_{1,2,3,w}$ performed by nine groups g of test subjects using the same values of the controlled variables of indoor light environment, the average group scores $T_{w,avg,L,g}$ and the standard deviation of the scores $s_{L,g}$ are shown (a); normalized test scores $T_{n,L,g}$ (b) and comparison of the normalized test scores $T_{n,L,g}$ of the presented experiment with the Kruthof relation (c).

for researchers in the field of indoor environment-quality design and experiment planning, as it is evident that white-noise exposure alone does not adequately predict indoor environmental quality in the noise-exposure domain (IEQ_N). Our results are also useful for the designers of natural/hybrid ventilation through openings in the building envelope that cause increased exposure to traffic noise (in urban environments). We conclude that both the sound-pressure level and the type of noise must be included as influencing variables in the prediction model for the

single-score productivity metric.

3.1.5. Interdependence of the variables of the IEQ on the performance of mental work

An important guideline when creating the integral indicators as a single-score metric for the productivity of mental work is that there is no correlation between the variables of state for the examined IEQ domains (thermal comfort, IAQ, light environment and noise exposure in our

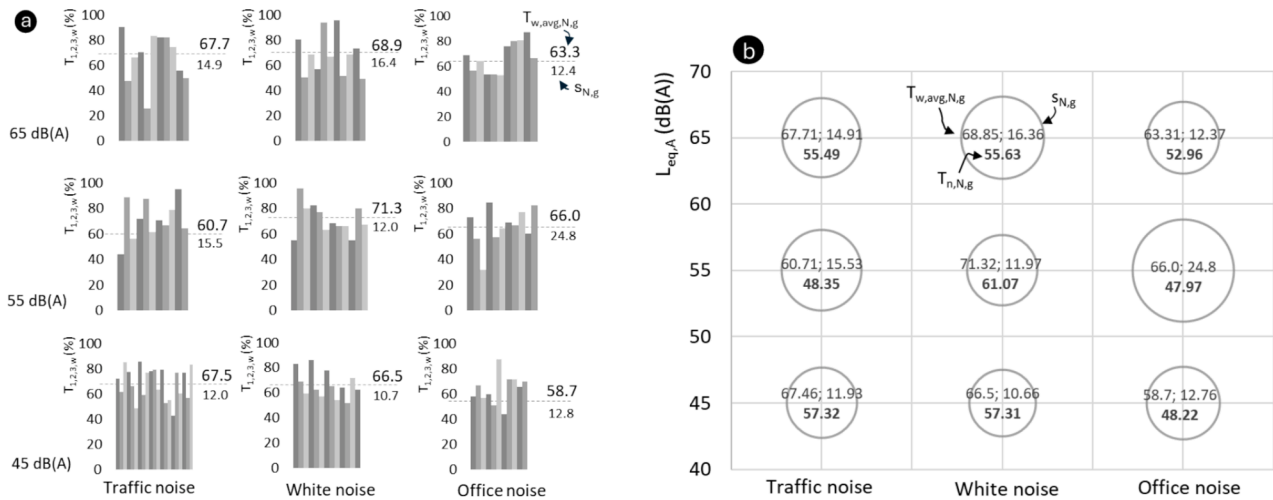


Fig. 11. Weighted test scores $T_{1,2,3,w}$ of nine groups g of test subjects who performed the test at the same values of the controlled variables of the noise exposure, the average group score of tests $T_{w,avg,N,g}$ and the standard deviation of the groups' score of tests $s_{N,g}$ (a) and the normalized average scores of tests $T_{n,N,g}$ (b).

study) according to the impact on the scores of tests. If this is fulfilled, an integral indicator can be developed in the simplest way by “summarizing” the indicators of the individual IEQ domain. To prove this, the statistical analysis of the correlation between the indoor environment variables, e.g., shown for test T1 (d2 test) in Fig. 12(a), was performed with the DATAtab software [45] for each test of mental work (T_1 , T_2 , T_3). The results are shown in Fig. 12(b). The Spearman test shows that there is no ($r < 0.1$) or only a low ($0.1 \leq r < 0.3$) correlation between the scores of the tests and any of the measured/controlled variables of the IEQ domains, and that the statistical significance of the correlation is $p \gg 0.05$. Together with the presented experimental results, evaluated

with normalized average scores of tests T_n , this finding will be the basis for developing the single-score productivity metric as the productivity-loss index $PL-IEQ_{INDEX}$. The index will predict how the productivity of mental work decreases if the indoor environment conditions differ from the ideal ones at which the highest normalized average scores of tests T_n were achieved by the test subjects.

4. Integral single-score metrics of indoor environmental quality and the productivity of mental work

In Section 3 we showed that despite the indoor environment

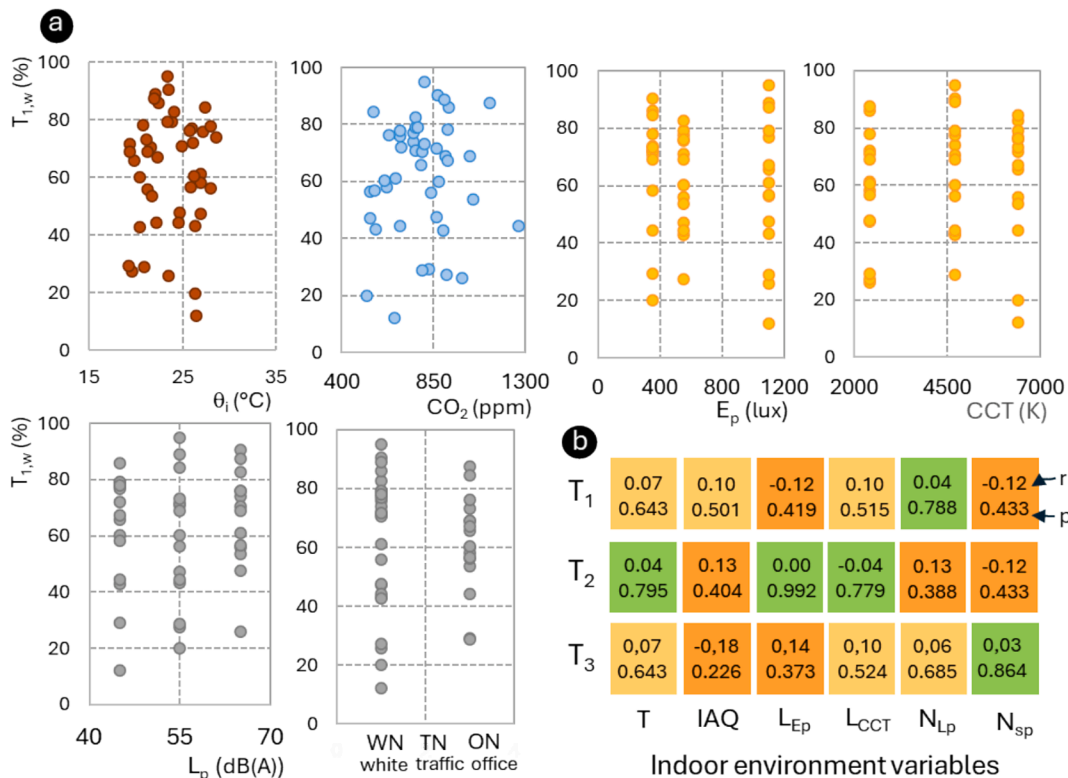


Fig. 12. Weighted scores of tests T_1 (d2 test) achieved by the test subjects exposed to a range of physical variables of IEQ domains in the IEQLab during the experiment (a); Spearman's correlation (r) and statistical significance of the correlation (p) between the weighted scores of tests $T_{1,2,3,w}$ and the variables of the indoor environment (b).

variables being maintained within the limits of the standardized IEQ quality classes, the indoor environment conditions affect the productivity of mental work. In the following section two single-score indices were developed to summarize the impact of the indoor environment's physical variables on the productivity of mental work: the index of indoor environmental quality (IEQ_{INDEX}) and the productivity-loss index ($PL-IEQ_{INDEX}$).

4.1. Productivity-loss index – $PL-IEQ_{INDEX}$

With the productivity-loss index $PL-IEQ_{INDEX}$ the decrease in performance of mental work is measured with a relative approach according to the highest productivity that can be achieved under the best conditions for each IEQ domain, i.e., thermal comfort, indoor air quality, lighting environment and noise exposure.

The starting point for the development of the $PL-IEQ_{INDEX}$ is the experimental normalized scores of the tests of mental work performance $T_{n,(T,IAQ,L,N)}$ (%) presented in Section 3 (shown in Figs. 8–11). Note that the subscript (T,IAQ,L,N) elsewhere means T or IAQ or L or N . The following steps were taken to develop the $PL-IEQ_{INDEX}$:

- For each IEQ domain the conditions in which the normalized scores of the tests $T_{n,(T,IAQ,L,N)}$ are the highest are set as the reference (Fig. 13, (1)) and the productivity loss for these indoor conditions is set to zero ($PL_{ref} = 0$ %). The differences in the productivity of mental work ΔPL (%) between the reference and all the other $T_{n,(T,IAQ,L,N)}$ within IEQ domain are determined (Fig. 13, (2)).
- The maximum difference in productivity loss ΔPL_{max} found among all the IEQ domains defines the scale of the index (Fig. 13, (3)).
- The values of the productivity loss ΔPL were converted into the productivity of the mental work ranks $R_{PL,(T,IAQ,L,N)}$ using the scale recommended in [46] with the values $C_{ref} = 1.5$ and $n_{ref} = 5$, where C_{ref} represents the nonlinearity of the scale and n_{ref} is the number of ranks (Fig. 13, (4)):

$$R_{PL,(T,IAQ,L,N)} = \sqrt{C_{ref}}^{(n-n_{ref})}$$

By implementing the proposed procedure for the determination of the productivity ranks $R_{PL,(T,IAQ,L,N)}$, the productivity loss and ranks for each IEQ domain were determined and the results are presented in Fig. 14. The productivity losses, ΔPL , are shown for each of the assessed IEQ domains, together with ranks of the productivity of mental work R_{PL} according to the scale shown in Fig. 13 (4). These are the most favoured indoor environment conditions at which $\Delta PL = 0$ can be found as an indoor environmental quality class (for thermal comfort and IAQ) or as discrete values of a pair of physical variables (for light environment and noise exposure). In Fig. 14 the standard deviation of the scores of the tests $s_{(T,IAQ,L,N)}$ are shown on a scale of circle diameters and it can be confirmed that under the most favoured conditions, the standard deviation is the smallest. This indicates that those conditions will be optimal for performing mental work for most of the building occupancies.

By defining the ranks $R_{PL,(T,IAQ,L,N)}$ of the productivity loss for each IEQ domain, the impact of the physical variables of the indoor environment on the productivity of the mental work can be “summarized” in a productivity index $PL-IEQ_{INDEX}$, as an integral, single-score metric. The weighted Euclidean-distance (UER) method [47,48] was used. The method assumes that the impacts of the variables are independent, which was proven in Section 3.1.5 and Fig. 12. The positive Euclidean distance (UER^+) (Eq. (2)) represents the sum of the differences between the maximum rank (equal to 5) and the actual value of the rank of the productivity $R_{PL,T,IAQ,L,N}$ obtained for each domain at each time step of the numerical modelling (e.g., each hour during the year) in the BIM tool. Similarly, the negative Euclidean difference (UER^-) (Eq. (3)) represents the sum of the differences between the actual value of the rank of the productivity $R_{PL,T,IAQ,L,N}$ and the minimum rank (equal to 1). The following expressions were used:

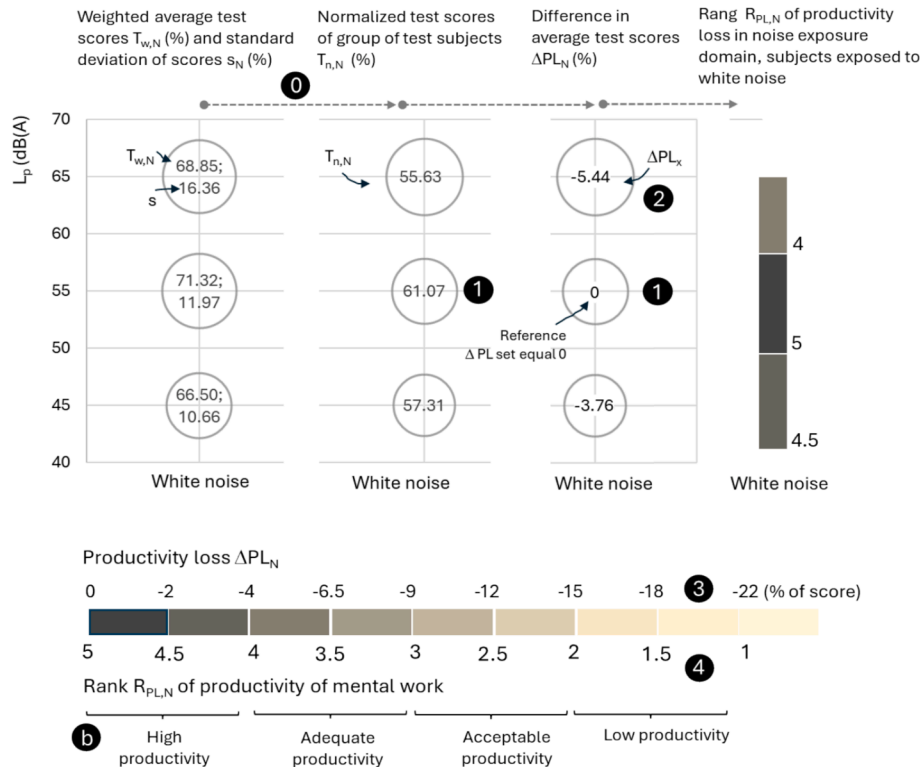


Fig. 13. Steps taken to develop specific IEQ domain productivity rank $R_{PL,(T,IAQ,L,N)}$ shown for $R_{PL,N}$ when building occupancies are exposed to white noise; descriptive values of productivity of mental work suggested according to the productivity rank are shown (b).

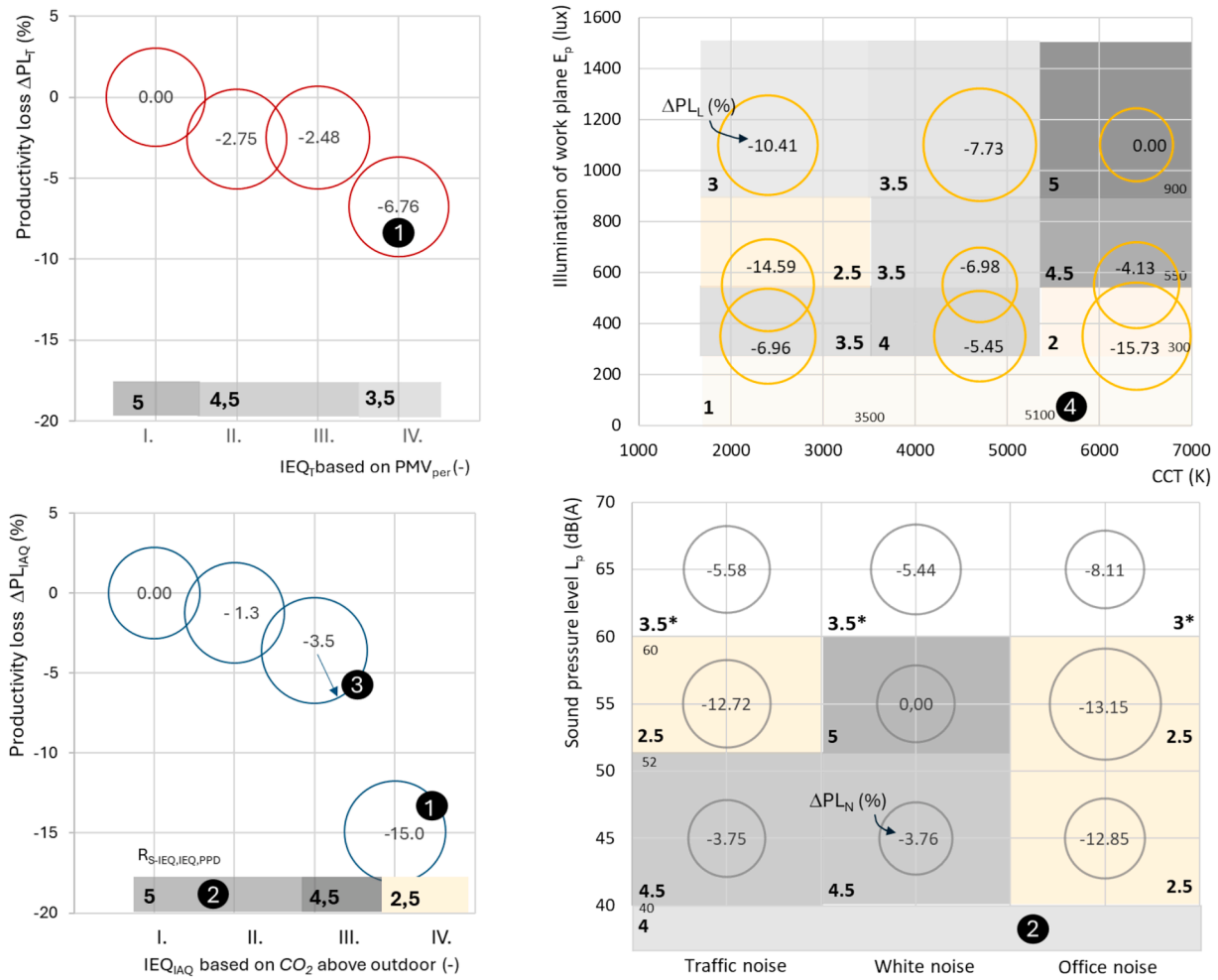


Fig. 14. Productivity loss of mental work performance $\Delta PL_{(T,IAQ,L,N)}$ (1) under different indoor environment conditions in the assessed IEQ domains, ranks of productivity of mental work $R_{PL_{(T,IAQ,L,N)}}$ (2) and standard deviations of test scores $s_{(T,IAQ,L,N)}$ (3) visually presented with the diameter of the circles; the $R_{PL_{(T,IAQ,L,N)}}$ rank's boundary values of the physical variables that will be used in dynamic modelling are shown (4). Note: * Values are informative.

$$UER_o^+ = \sum_{j=1}^4 z_k \cdot \left(\overbrace{R_{PL,max}}^5 - R_{PL,j} \right) (-) \quad (2)$$

$$UER_o^- = \sum_{j=1}^4 z_j \cdot \left(R_{PL,j} - \overbrace{R_{PL,min}}^1 \right) (-) \quad (3)$$

$$PL - IEQ_{INDEX,t} = \frac{UER_o^-}{UER_o^- + UER_o^+} \cdot 100 (\%); \quad 0 \leq PL - IEQ_{INDEX,t} \leq 100 \quad (4)$$

$$PL - IEQ_{INDEX} = \frac{\sum_{t=1}^T PL - IEQ_{INDEX,t}}{\frac{T}{\Delta t}} (\%) \quad (5)$$

where j represents the IEQ domain (T, IAQ, L, N), while t indicates that the index (Eq. (4)) is evaluated for each time step Δt within the observation period T of the building response modelling. T can be a day, the coldest week, a month of the year or an annual value ($T = 8760$ h). By averaging the $PL - IEQ_{INDEX,t}$ during the time period T , the integral single-score productivity-loss index $PL - IEQ_{INDEX}$ (Eq. (5)) is determined.

4.1.1. Weighting factors of the IEQ domains' productivity-loss index

To adjust the impact of the individual IEQ domain on the productivity-loss index, weighting factors $z_{(T,IAQ,L,N)}$ were introduced. The value of z is between 0 and 1, while $\sum z(T,IAQ,L,N) = 1$.

In the literature [3,47,49] the following values for the weighting factors can be found: z_T between 0.12 and 0.42 for the thermal comfort domain, z_{IAQ} between 0.14 and 0.36 for the IAQ domain, z_L 0.16 to 0.25 for the light environment and z_N 0.16 to 0.39 for the noise exposure IEQ domain. Because a wide range of values were published, we proposed a new approach for determining the weighting factors based exclusively on the standard deviation of the scores for tests of mental work performance. We assumed that the most important domain is the one in which the scores for the tests achieved by the test subjects have the largest standard deviation:

$$z_j = \frac{s_j}{\sum_{j=1}^4 s_j} (-) \text{ with } \sum_{j=1}^4 z_j = 1, \quad (6)$$

where j indicates the assessed IEQ domain. According to our study, the standard deviation of the scores for the tests were $s_T = 12.36$, $s_{IAQ} = 13.70$, $s_L = 14.59$ and $s_N = 15.50$, which corresponds (Eq. (6)) to the weights $z_T = 0.220$, $z_{IAQ} = 0.244$, $z_L = 0.260$ and $z_N = 0.276$. The largest number indicates the IEQ domain with largest relative impact on the productivity of mental work (noise exposure according to our research), but not necessarily with the largest impact on the productivity index $PL - IEQ_{INDEX}$, because the index is also related to the normalized score of the tests.

4.2. Indoor environmental quality index – IEQ_{INDEX}

The assessment of the productivity of mental work with $PL-IEQ_{INDEX}$ presented in Section 4.1 is based on physical variables determined by the dynamic modelling of a building's response (e.g., θ_i , θ_{op} , CO_2 , E_p , L_p) or designed by the building planner (*CCT*, *spectrum/type of noise*, *definition of specific outdoor conditions*).

Therefore, it will be useful for planners to be able to evaluate a building's design simultaneously with an integral single-score metric of indoor environmental quality based on physical variables ranked in standardized classes I to IV. For this, the indoor environmental quality index IEQ_{INDEX} was developed, as presented in this study. To provide building planners with an even more detailed overview of the indoor environment, an extended number of indicators was included in the assessment of IEQ. The list and ranking of the indicators included in the evaluation of the IEQ_{INDEX} are shown in Table 1. The list is based on standardized values, literature suggestions and our research data. The developed algorithm for the determination of IEQ_{INDEX} allows building designers to freely select indicators that they want to include in the evaluation.

Data from a building's dynamic response modelling are used to determine the indoor environmental quality index IEQ_{INDEX} . The following procedure, based on the durations of the quality classes of the IEQ indicators, was developed:

- At each time t within the observation period T defined by the building planner (day, week, ... year), the quality class of each of k -th selected IEQ indicator is determined as $IEQ_{(I),k,t}$ to $IEQ_{(IV),k,t}$.
- Assuming that dynamic modelling is performed in constant time steps Δt (usually 1 h), the number of samples $N_{IEQ(I-IV),k}$ in each of IEQ quality class (I-IV) for each IEQ indicator (k) is counted over the observation period (Eq. (7)):

$$N_{IEQ(I),k} = \sum_{t=1}^{\frac{T}{\Delta t}} n_{IEQ(I),k,t}, \quad N_{IEQ(II),k} = \sum_{t=1}^{\frac{T}{\Delta t}} n_{IEQ(II),k,t} \dots \text{and } N_{IEQ(I),k} + N_{IEQ(II),k} + N_{IEQ(III),k} + N_{IEQ(IV),k} = \frac{T}{\Delta t} \quad (7)$$

- The relative duration $D_{IEQ(I-IV),k}$ of IEQ quality class of each IEQ indicator is calculated (Eq. (8))

Table 1

List of physical variables as indicators of IEQ included in the evaluation of IEQ_{INDEX} in this study; orange fields indicate indicators that were included in the case-study example (Section 6).

IEQ domain	IEQ domain indicators	Source	IEQ Class I	Class II	Class III	Class IV
Thermal comfort	Operative temperature θ_{op} (°C)	Standardized value ^a heating period	21–23	20–24	19–25	17–25
		Standardized value ^a cooling period	23.5–25.5	23–26	22–27.5	21–28
	Adaptive operative temperature $\theta_{op, adap}$ (°C)*	Standardized value ^b heating, cooling period	Same IEQ class ranges as by non-adapted operative temperature assessment.			
	Humidity Index HI (–)	Literature ^c	60–65	65–70	70–80	> 80
	PMV (Fanger model)	Standardized value ^a	± 0.2	± 0.5	± 0.7	> ± 0.7
	Personal PMV_{per} (–)	Experiment, heating period	–1.2 to –0.7	–1.575 to 0.325	–1.825 to 0.075	< –1.825 and > 0.075
IAQ		Experiment, cooling period	–0.56 to –0.06	–0.935 to 0.315	–1.185 to 0.565	< –1.185 and > 0.565
	CO_2 (ppm above out.)	Standardized value ^a	< 350	< 500	< 800	≥ 800
	Predicted percentage of dissatisfied PD (%)	Standardized value ^a	< 15	< 20	< 30	≥ 30
	Decipols (dp)	Designer's choice	≥ 1 dp (10 l/s per)	> 0.75 dp (7 l/s per)	≥ 0.4 dp (4 l/s per)	< 0.4 dp
	Lighting comfort					
	Illumination (lux)	Standardized value ^a	750–1500	> 550	> 350	< 350 or > 1500**
Noise exposure	DGP Daylight Glare Probability (–)***	Dynamic modelled	< 0.35	0.35–0.40	> 0.4–0.45	> 0.4
	Melanotropic luminous flux EML (lux)****	Designer's choice (<i>CCT</i>), dynamic modelled (daylight)	> 250	> 200	> 150	< 100
	Sound pressure level $L_{eq,A}$ (dB (A)), sound spectrum	Standardized value ^a ; reduced façade sound retention at open window*****; Liter. ^d	≤ 35	≤ 40	≤ 45	> 45

Notes: a [4], b [50], c [23], d [30].

* Adaptive operative temperature quality class was adopted from Norme SIA 380/1-C1:2019 [50] based on 48 h sliding average of outdoor temperature. $\theta_{e,48}$ is chosen by designer (in our study set to 15 °C). Model incorporates seasonal adaptation of set point operative temperature as well as conditions of free running buildings (e.g. natural cooled). Range of classes are the same as by non-adapted operative temperature.

** Glare avoidance.

*** DGP (Daylight Glare Probability) is assess according to high light flux emitted from sun or indoor surfaces seen by observer in the central direction of view, usually at the incident angle bellow 20° in the direction of view. Procedure for determination and range of DGP quality classes were taken from [51]; constants in simplified DGP model were taken from [52] as $a = 6.22 \cdot 10^{-5}$, $b = 0$ in $c = 0.184$.

**** Melanotropic luminous flux received by the face of a seating person is determined by the melanopic ration and depends on the correlated light colour *CCT*. EML is calculated according to literature [53–55].

***** The impact of the façade ventilation openings on increased indoor sound pressure is modelled according to ration of opening and window area and sound resistance of the windows according to the experimental results presented in [56].

$$D_{IEQ(I),k} = \frac{N_{IEQ(I),k}}{\frac{T}{\Delta t}}, D_{IEQ(II),k} = \frac{N_{IEQ(II),k}}{\frac{T}{\Delta t}} \dots \text{and } D_{IEQ(I),k} + D_{IEQ(II),k} + D_{IEQ(III),k} + D_{IEQ(IV),k} = 1 \quad (8)$$

- In the last computational step, the single-score, indoor environmental quality index IEQ_{INDEX} is determined (Eq. (9)) as follows:

$$IEQ_{INDEX} = w_I \cdot \left(\sum_{j=1}^4 v_j \cdot \sum_{k=1}^m u_k \cdot D_{IEQ(I),j,k} \right) + w_{II} \cdot \left(\sum_{j=1}^4 v_j \cdot \sum_{k=1}^m u_k \cdot D_{IEQ(II),j,k} \right) + w_{III} \cdot \left(\sum_{j=1}^4 v_j \cdot \sum_{k=1}^m u_k \cdot D_{IEQ(III),j,k} \right) + w_{IV} \cdot \left(\sum_{j=1}^4 v_j \cdot \sum_{k=1}^m u_k \cdot D_{IEQ(IV),j,k} \right), \quad (9)$$

where j represents the assessed IEQ domain (T, IAQ, L, N), k is the IEQ indicator in j -th IEQ domain, while u_k , v_j and w_{I-IV} are weighting factors. u_k weighted the impact of the k -th IEQ indicator within each of the

IEQ domains, while v_j weighted the impact of IEQ domains on IEQ_{INDEX} . u_k and v_j can be selected by the building planner while keeping $\sum u_k$ and $\sum v_j$ equal to 1. If, for example, all three IEQ indicators of the thermal comfort domain and all four IEQ domains are assumed to be equally important for building occupancies, then $u_1 = u_2 = u_3 = 0.333$ and $v_T = v_{IAQ} = v_L = v_N = 0.25$ should be inserted into the IEQ_{INDEX} evaluation custom-developed macro in the BIM/BEM tool. However, we could suggest values of v_j determined according to the performance of the mental work presented in Section 4.1 – $v_T = 0.220$, $v_{IAQ} = 0.244$, $v_L = 0.260$ and $v_N = 0.276$. The weighting factors w_{I-IV} of the IEQ classes can also be chosen, although we recommended factors suggested in [57,58]: $w_I = 100$, $w_{II} = 75$, $w_{III} = 35$ and $w_{IV} = 0$. Accordingly, the IEQ_{INDEX} has a scale between 0 and 100.

5. Integration of PL-IEQ_{INDEX} and IEQ_{INDEX} in BIM tool

The productivity-loss index PL-IEQ_{INDEX} and the indoor environmental quality index IEQ_{INDEX} are very dependent on the unsteady state of the outdoor climatic conditions, unsteady heat transfer and adaptive properties of building envelope structures and the schedules for a building's use. Therefore, the dynamic modelling of a building's response is needed to predict both. Several building-information modelling (BIM) and building-energy modelling (BEM) tools are

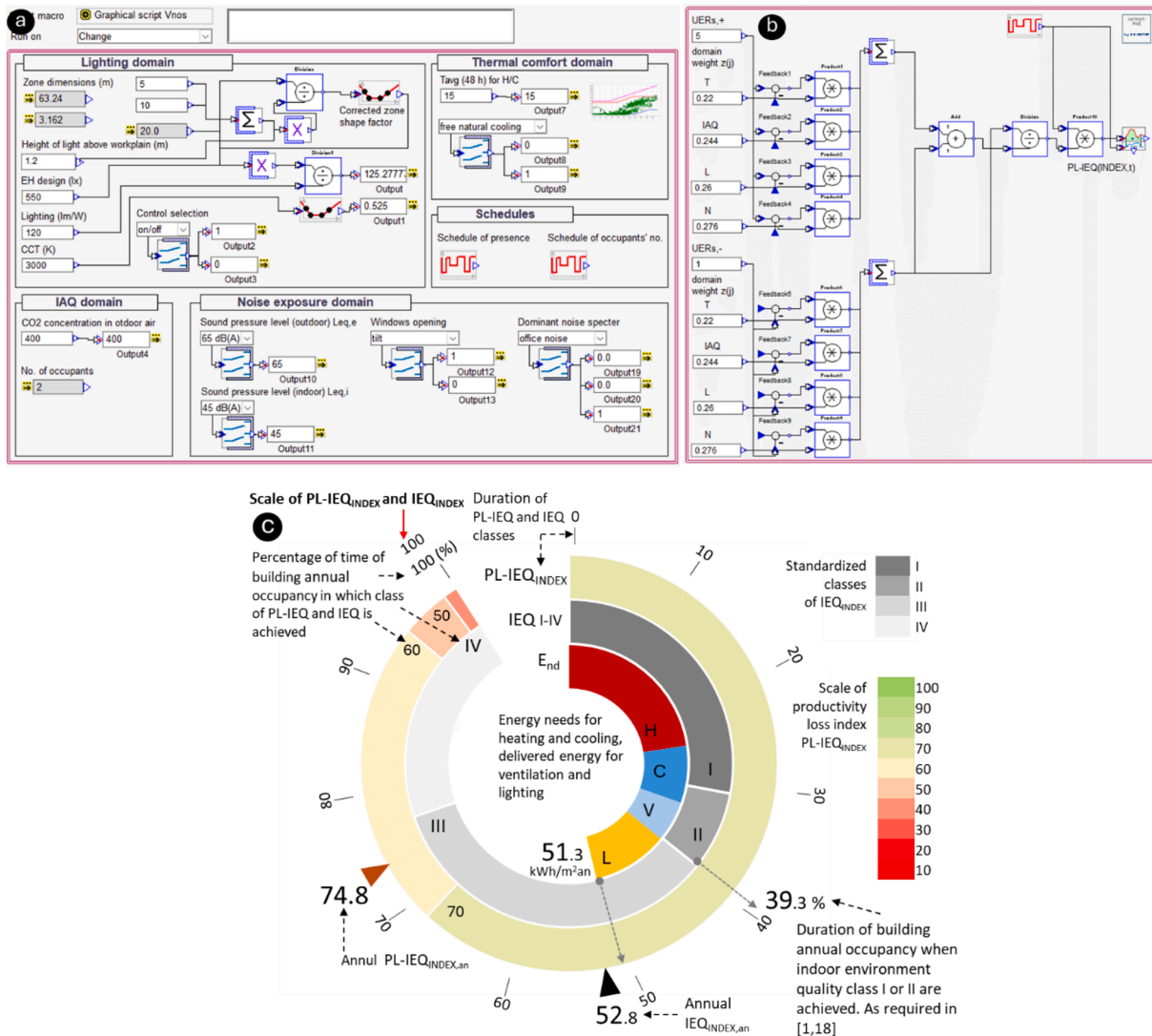


Fig. 15. Print screen of graphical scripts for user-defined data inputs (a), macro for determination of Euclidean distances (b) and example of possible visualization of modelled annual data with upgraded BIM/BEM tool (c).

available to building designers, and one of those, IDA-ICE [59], was used in our study. As in the case of other BIM/BEM tools, the integral indoor-quality assessment is not available by default (except the standardized *PMV*) and must be built into the software tools. In our case, this was done using a graphical programming technique by developing graphical scripts and macros. This simple way of programming has the advantage of a simple, user-friendly definition of the input data, a description of the computational algorithms, and the presentation of results in an easy-to-present manner. But probably the most welcome feature is the possibility to simply exchange the macros with other users (of IDA-ICE tool). Scripts and macros are connected to available built-in IDA-ICE modules and therefore no additional adaptation of the building model is needed, and all the data for energy demand in the buildings are read in developed results-presentation macro. In Fig. 15, examples of the graphical script for user-defined data inputs (a), the macro for a determination of

Euclidean distances (UER_0^+ and UER_0^-) (b), and one of the possible presentations of BIM/BEM (c) modelled annual data are shown.

6. Case study: relations between climate, energy demand, $PL-IEQ_{INDEX}$ and IEQ_{INDEX}

For a demonstration of dynamic modelling with scripts and macros an upgraded IDA-ICE BEM tool, office with four workplaces occupied 3640 h per year, with double side windows (in south and north facades) and a useful area of 50 m² was analysed. The boundary conditions are presented in Annex 1. To evaluate the impact of the energy performance of the case-study space on the $PL-IEQ_{INDEX}$ and IEQ_{INDEX} , three levels were compared: “pre-EPBD building”, “nearly-zero-energy building” (current EPBD required nearly-zero-energy building) and building in which the advanced control algorithms of technical building systems are

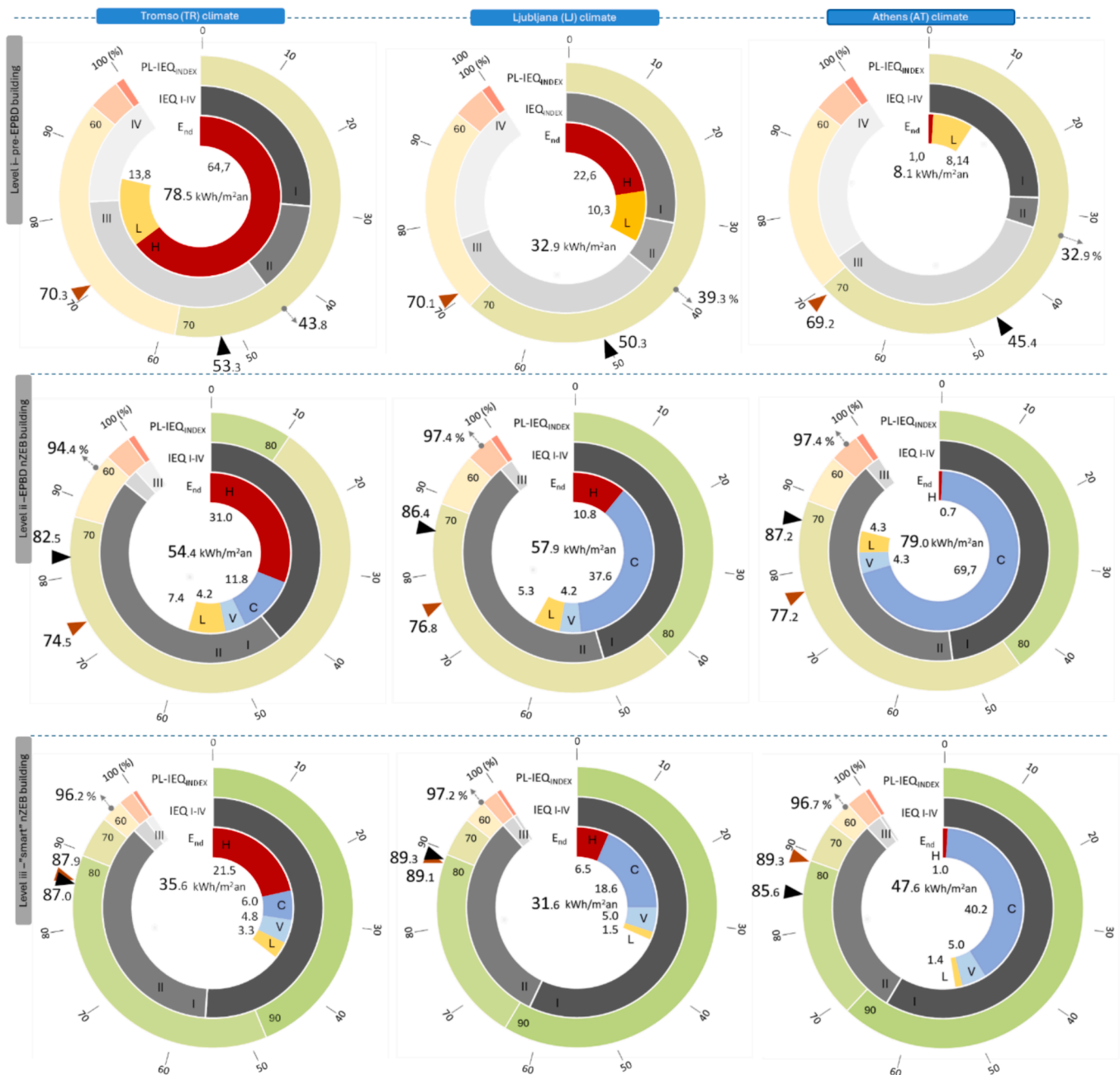


Fig. 16. Annual $IEQ_{INDEX,an}$, $PL-IEQ_{INDEX,an}$ and energy demand for heating (H) and cooling (C) (as energy needs) and mechanical ventilation (V) and lighting (L) (as delivered energy) of case-study office space.

utilized (nearly-zero-energy building with a high smart-readiness indicator SRI [5]). All the cases were studied for three climate conditions: Tromsø (TR; Koppen classification Dfc), Ljubljana (LJ, Cfb) and Athens (AT, Csa). Meteorological data in form of a Typical Meteorological Year were used. The energy demands are presented as energy needs for heating and cooling (implementing the ideal heater/cooler) and as electricity for ventilation and lighting (delivered energy) without the weighting of energy carriers. The results are shown in Fig. 16 in the way shown in Fig. 15(c).

From Fig. 16 we see that in the pre-EPBD building the required I and II IEQ class of selected IEQ indicators (Table 1) are only established in 32.9 % (AT), 39.3 % (LJ) and 43.8 % (TR) of the annual working hours. The IEQ_{INDEX} is in the range between 45.4 (AT) and 53.3 (TR), because of the lack of installed building services (mechanical ventilation and cooling) typical for pre-EPBD buildings. Unfavourable indoor environment conditions have slightly less impact on the productivity, nevertheless, the $PL-IEQ_{INDEX}$ is in the range of 70, meaning that productivity is not affected strongly by the poorer indoor environment conditions.

Introducing building technical systems to control indoor conditions in all the IEQ domains in nearly-zero energy-buildings causes an increase in the specific (per m^2) annual energy demand in warm and moderate climates, while the building envelope's renovation (Annex 1) only decreased the energy use in a cold climate (TR). Nevertheless, the number of working hours during which the required IEQ classes I and II increase (up to 94 % to 97 %) and the IEQ_{INDEX} increases to a pleasant 82.5 % (TR), 86.4 % (LJ) and 87.2 % (AT). The impact on productivity is not so pronounced, although the $PL-IEQ_{INDEX}$ rises by 4.1 % (TR), 6.7 % (LJ) and for 8 % (AT), while the figures are below the IEQ_{INDEX} .

At the third level, a nearly-zero-energy building was upgraded with algorithms for the adaptive operation of building-service systems as well as with adaptive algorithms for the utilization of environmental sources of heating, cooling and lighting adjusted to local climatic conditions (Annex 1) and occupancies. A huge impact on the annual energy demand can be seen in all climates, since the specific energy demand decreases by 35 %–45 % compared to a “non-smart” nearly-zero-energy building. The IEQ_{INDEX} increases by an additional 3 %–5 % (except for Athens as consequence of extensive night cooling with ventilation) while the $PL-IEQ_{INDEX}$ increases even more, 12.1 % up to 13.4 % and both indexes are now approximately equal.

The presented case is developed only for a typical office space, but evidently shows the necessity for the simultaneous observation of diverse indicators of indoor environmental quality over mere energy demand.

7. Conclusions

The paper consists of four connected research topics. (1) Developing a procedure for ranking productivity loss based on the experimental data of three types of tests of mental work in office-like indoor environments in four assessment IEQ domains, i.e., thermal comfort, IAQ, lighting environment and noise exposure. (2) Developing and demonstrating a method for an integrated single-score assessment of the productivity of mental work with a productivity-loss index $PL-IEQ_{INDEX}$. An index value 100 points means that optimal conditions for performing mental work are ensured during all occupancy periods and in all IEQ domains. To further support building planners during decision making, the indoor environmental quality index IEQ_{INDEX} as an integral single-score assessment of IEQ based on the standardized ranking of IEQ (classes I–IV) was developed and an index value of 100 means that the top IEQ class (I) is ensured during the all the occupancy hours and in all the IEQ domains during the observation period T. (3) As indices depend on a building's dynamic response, determination procedures were integrated in the BIM/BEM tool by custom-developed graphical scripts and macros, which are linked to calculation algorithms in the BEM tool and can be, as a custom module, transferred to any building project or among the building planners (of used BIM/BEM IDA-ICE tool). (4) The case study

confirmed that a building's evaluation should not be based solely on an energy-performance evaluation, but simultaneously on the evaluation of two other integrated metrics: living comfort and productivity.

While the results of the research are discussed in detail, following major conclusions can be drawn:

- Even when maintaining the required range of standardized indoor environmental quality classes, the productivity of mental work decreases from top ranked I class down to IVth. In the case of combined indicators (light environment and noise exposure) for which standardized IEQ classes do not exist, the ranking of productivity of mental work is proposed.
- We prove that tests of mental work widely used in the published research represent very different mental workloads and we propose a weight to compensate for this in the procedure of developing a productivity-loss index and findings can serve researchers while designing the tests of mental work.
- The ranking of the productivity of mental work is based on normalized scores of tests, which include average values, as well as scores of standard deviations, which we believe reveal the personal characteristics of test subjects better and summarize their response.
- We did not find statistically significant differences in the scores of tests of mental work regarding to the gender and age of the test subjects.
- With statistical methods to evaluate the experimental results of the productivity of mental work it was proven that productivity could be assessed separately for each of the IEQ domains and afterwards summarized in a productivity-loss index $PL-IEQ_{INDEX}$. The same assumption, which simplified the determination of single score indexes, was used for defining the IEQ_{INDEX} .
- The weights of the impact of each of the four IEQ domains on the $PL-IEQ_{INDEX}$ were determined only on the basis of standard deviation of experimental data, showing that noise exposure and lighting IEQ domains are the most influential, although the difference among the weights of the domains are in the range of ± 10 %;
- The developed algorithms for ranking of IEQ domain variables, the determination of $PL-IEQ_{INDEX}$ and IEQ_{INDEX} , are based on a physical variable obtained by the dynamic modelling of the building response and can therefore be used in combination with the gathered data from any of the established dynamic building-modelling tools, while examples of custom-developed graphical scripts and macros in selected BIM/BEM tool (IDA-ICE) are presented.
- The case study confirms that indoor environmental quality increases when implementing energy-efficiency measures, while the smart operation of a building's technical system is needed to establish high productivity of mental work as well. Saying that, the multiple-perspectives approach to building-design assessment presented in the study is justified.

The limitation of the study could be the narrow range of indoor environment variables. However, this could also be advantage of the study, because we did not force the test subjects to undertake experiments in conditions that should not appear in buildings. Limitations could be that the tests of mental work were performed in an “office-like” IEQLab with an artificial window, and that the mental work was relatively short-term and intense. This would be different to some extent in a real work environment. Nevertheless, the results of the study can be presented as a benchmark for further research in real office environment. The productivity was only studied for mental work performed by setting person, although such human activity is quite common even in residential buildings. Although the experiments were conducted half in the heating and half in the cooling season, it will be useful that in further research, seasonal adaptations and cross-relationships will be studied in more detail. In the future an additional index of perceived indoor environmental quality could be introduced. Nevertheless, we see the presented study as an introduction step in the BIM process.

Declaration

The research was approved by the University of Ljubljana's Commission for Ethics in Research Involving Work with Human Subjects, No. 007-2021.

CRediT authorship contribution statement

Suzana Domjan: Writing – original draft, Validation, Formal analysis, Data curation, Conceptualization. **Rok Fink:** Writing – review & editing, Formal analysis. **Sašo Medved:** Writing – review & editing, Supervision, Conceptualization.

Appendix 1

Boundary conditions and control variables for case study building/office space.

Level i – pre-EPBD building

Office space with south and north orientated window; $A_{use} = 50 \text{ m}^2$, $V = 150 \text{ m}^3$
Location: Tromsø (TR, Φ 69.64 N), Ljubljana (LJ, Φ 46.05 N), Athens (AT, Φ 37.98 N), build urban environment, outdoor noise pressure level during occupancy period $L_{eq,e,dn} = 55 \text{ dB(A)}$.
Occupants: 4 people, 1.2 met, $clo = 0.85 \pm 0.2$, presence 7:00–17:00 (working days only); infiltration $n_{inf} = 0.2 \text{ h}^{-1}$, other internal heat gains 3.2 W/m^2 ; office noise, designed indoor sound pressure level 45 dB(A) (closed windows).

Thermal envelope: thermal transmittance opaque structures $U = 0.146 \text{ W/m}^2\text{K}$, thermal transmittance of transparent structures $U_w = 1.584 \text{ W/m}^2\text{K}$, $g = 0.479$, $\tau_{vis} = 0.696$, orientation south + north, window-to-wall ration = 0.667 , $A_w/A_{use} = 0.20$, internal shading $\tau_{sol} = 0.3$, $\tau_{vis} = 0.55$.

Technical buildings systems: natural ventilation (infiltration + window opening), electrical light source efficiency 60 lm/W , $CCT = 3000 \text{ K}$.

Control algorithms: setpoint temperature $\theta_{i,set}$ heating period $20.5^\circ\text{C}/23.5^\circ\text{C}$, changeover between heating and cooling at 48 h sliding average outdoor air temperature $\theta_{e,48} = 15^\circ\text{C}$, shading ON at 100 W/m^2 on the facade, assumed opening of window for natural ventilation by users ($1400 \pm 200 \text{ ppm}$, 100% of window height, 10% window width), luminaries ON $E_p < 350 \text{ lx}$, OFF $E_p > 500 \text{ lx}$ at work plane.

Level ii – EPBD nZEB building¹

Location and occupancy: as Level i; infiltration $n_{inf} = 0.02 \text{ h}^{-1}$, other internal heat gains 3.2 W/m^2 ; reduced office noise, designed indoor sound pressure level 40 dB(A) .

Thermal envelope: thermal transmittance opaque structures $U = 0.146 \text{ W/m}^2\text{K}$, thermal transmittance of transparent structures $U_w = 1.000 \text{ W/m}^2\text{K}$, $g = 0.533$, $\tau_{vis} = 0.694$, orientation south + north, window/wall ration = 0.667 , $A_w/A_{use} = 0.20$, external shading $\tau_{sol} = 0.2$, $\tau_{vis} = 0.55$.

Technical buildings systems: mechanical ventilation CAV $V_v = 160 \text{ m}^3/\text{h}$ ($40 \text{ m}^3/(\text{h.pers.})$), specific fan power $SFP_{sup} = 0.75 \text{ kW}/(\text{m}^3/\text{s})$, $SFP_{extr} = 0.5 \text{ kW}/(\text{m}^3/\text{s})$, mechanical cooling, infiltration 0.02 h^{-1} , luminaires 120 lm/W , $CCT = 4700 \text{ K}$.

Control algorithms: setpoint temperature $\theta_{i,set,H}$ heating period $20.5^\circ\text{C}/23.5^\circ\text{C}$, setpoint for cooling $\theta_{i,set,C}$ $23.5^\circ\text{C}/25.5^\circ\text{C}$, changeover between heating and cooling at 48 h sliding average outdoor air temperature $\theta_{e,48} = 15^\circ\text{C}$, shading ON at $G_{glob,90} 100 \text{ W/m}^2$ on the facade, luminaries ON $E_p < 350 \text{ lx}$, OFF $E_p > 500 \text{ lx}$ at work plane.

Level iii – “smart” nZEB building

Location and occupancy: as Level i; outdoor noise pressure level during occupancy period $L_{eq,e,dn} = 45 \text{ dB(A)}$; masked indoor noise – white noise at sound pressure level 40 dB(A) (closed windows).

Thermal envelope: optimized regarding to the location; thermal transmittance opaque structures $U = 0.146 \text{ W/m}^2\text{K}$, thermal transmittance of transparent structures TR: $U_w = 0.607 \text{ W/m}^2\text{K}$, $g = 0.601$, $\tau_{vis} = 0.768$; LJ $U_w = 0.583 \text{ W/m}^2\text{K}$, $g = 0.434$, $\tau_{vis} = 0.650$; AT $U_w = 0.583 \text{ W/m}^2\text{K}$, $g = 0.267$, $\tau_{vis} = 0.553$, orientation south + north, window/wall ration = 0.667 , $A_w/A_{use} = 0.20$, external shading $\tau_{sol} = 0.2$, $\tau_{vis} = 0.3$.

Technical buildings systems: mechanical ventilation with variable flow VAV $V_{v,max} = 288 \text{ m}^3/\text{h}$ (max $72 \text{ m}^3/(\text{h.pers.})$), specific fan power $SFP_{sup} = 0.75 \text{ kW}/(\text{m}^3/\text{s})$, $SFP_{extr} = 0.5 \text{ kW}/(\text{m}^3/\text{s})$, mechanical cooling, infiltration 0.02 h^{-1} , luminaires 140 lm/W , $CCT = 4700 \text{ K}$.

Smart Control algorithms: setpoint $\theta_{i,set,H}$ $20.5^\circ\text{C}/23.5^\circ\text{C}$, setpoint $\theta_{i,set,C}$ $23.5^\circ\text{C}/25.5^\circ\text{C}$, operative temperature, changeover between heating and cooling at 48 h sliding average of outdoor air temperature $\theta_{e,48}$, thermal comfort adaptive regarding top the site $\theta_{e,48}$: **A**² TR $\theta_{e,48} = 8^\circ\text{C}$, LJ $\theta_{e,48} = 12^\circ\text{C}$, AT $\theta_{e,48} = 14^\circ\text{C}$; **B**² shading ON at $\theta_i > 22.5^\circ\text{C}$ (overheating protection) or incident angle of sunrays $i < 20^\circ$ south (glare protection), shading ON at 100 W/m^2 on the facade (north); **C**² night-time natural cooling by ventilation at $\theta_i \geq 18^\circ\text{C}$, **D**² ventilation $\text{CO}_{2,max} 900 \pm 20 \text{ ppm}$ (II. class IAQ), **E**² dimmable LED ON $E_p < 500 \text{ lx}$ at work plane, daylight dependent, **F**² masking od office noise (into white noise).

Notes: ¹ national regulative SI ($E_{p,nren} < 75 \text{ kWh/m}^2\text{an}$); ² smart control algorithms.

Data availability

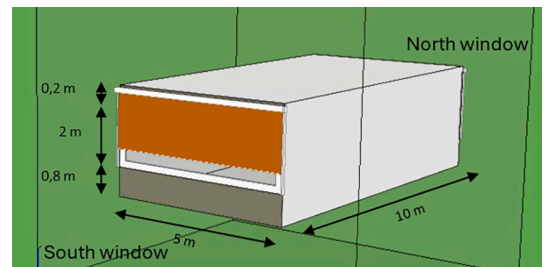
Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Suzana Domjan reports financial support was provided by Slovenian Research and Innovation Agency. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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